MONASH University

Semester Two 2015
Examination Period

Faculty of Engineering

EXAM CODES: **MAE3405**

TITLE OF PAPER: **FLIGHT VEHICLE PROPULSION PAPER 1**

EXAM DURATION: 3 hours writing time

READING TIME: 10 minutes

THIS PAPER IS FOR STUDENTS STUDYING AT: (Tick where applicable)

☐ Berwick	✓ Clayton	☐ Malaysia	☐ Off Campus Learning	☐ Open Learning
☐ Caulfield	☐ Gippsland	☐ Peninsula	☐ Enhancement Studies	☐ Sth Africa
☐ Parkville	☐ Other (specify)			

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No exam paper or other exam materials are to be removed from the room.

AUTHORISED MATERIALS

OPEN BOOK ☐ **YES** ✓ **NO**

CALCULATORS ✓ **YES** ☐ **NO**
(only calculators with an 'approved for use Faculty of Engineering' or 'Faculty of Science' sticker are permitted)

SPECIFICALLY PERMITTED ITEMS ☐ **YES** ✓ **NO**
If yes, items permitted are:

Candidates must complete this section if required to write answers within this paper

STUDENT ID: __ __ __ __ __ __ __ __ DESK NUMBER: __ __ __ __ __

Note the following: To gain full marks (**70**) you must: (a) correctly complete all three questions; (b) list assumptions used, provide the necessary diagrams and show all steps in your solutions; and (c) if you use an equation that does not appear on the equation sheet, show how the equation can be derived from those provided, or give a simple derivation of the equation, or show how it can be determined from the relevant thermodynamic cycle.

1. A single engine light aircraft is flying steady and level at 2000m with a true airspeed of 150kph. It is powered by a 4 cylinder, 4-stroke spark ignition piston engine whose data is tabulated below. If the engine operates on the fuel/air Otto cycle, **(a)** calculate the brake specific fuel consumption in units of kg/kWh and **(b)** estimate the drag acting on the aircraft and calculate the propulsive efficiency, list any assumptions. [15+10=25 Marks]

Quantity	Value	Quantity	Value
Compression ratio	8	Combustion efficiency	90%
Bore	0.1m	Mechanical Efficiency	85%
Stroke	0.12m	Volumetric efficiency	80%
RPM	2500	Equivalence ratio	1.0
Cp reactants	1.055kJ/kg-K	Cp products	1.486kJ/kg-K
Reactant mole weight	28.9kmol/kg	Product mole weight	28.0kmol/kg
Avgas Mol weight	114kmol/kg	Propeller Diameter	1.2m

2. The Russian Mil Mi-6 heavy lift helicopter is powered by two turboshaft engines with SSL rated power 4050kW each. Without a payload, its take-off mass is 28,000kg. The 5 bladed main rotor has a diameter 35m. If an amount equal to 20% of the SSL rated engine power is required to drive the tail rotor and aircraft accessories at any power setting, determine: **(a)** the maximum SSL level payload that can just be lifted from the ground and **(b)** the maximum altitude the helicopter can hover without a payload. [5+5=10 Marks]

3. A regional airliner has two, twin spool turbo-prop engines. Spool number one is the gas generator spool, while number 2 is the power spool, which drives the propeller through a gear box. For the engine specifications below, determine: **(a)** the thrust power of each engine and **(b)** the aircraft's fuel consumption in litres/hour. [30+5=35 Marks]

Quantity	Value	Quantity	Value
Compressor Ratio	12	Compressor Cp	1.005kJ/kg-K
Combustor F/A ratio	0.02	Compressor γ	1.4
Isentropic Efficiencies		Turbine Cp	1.148kJ/kg-K
- Compressor	85%	Turbine γ	1.333
- Turbine and power turb.	90%	Exhaust gas constant R	0.287kJ/kg-K
- Intake (subsonic)	95%	Intake nozzle area	0.2m^2
- Propelling nozzle	95%	Gear box efficiency	85%
Gas generator spool eff.	99%	Fuel	Jet A1
Combustor efficiency	98%	Altitude	7000m
Combustor pressure drop	3%	Speed	Mach 0.6
Propelling noz. press ratio	1.3	Propulsive efficiency	80%

END OF EXAM

Standard Atmosphere

Altitude Km	Temperature K	Pressure Pa	Density kg/m^3	Speed of Sound m/s
0	288.15	101325	1.225	340.29
1	281.65	89874.5	1.111	336.43
2	275.15	79495.2	1.006	332.52
3	268.65	70108.5	0.9091	328.57
4	262.15	61640.2	0.8191	324.57
5	255.65	54019.9	0.7361	320.52
6	249.15	47181.0	0.6597	316.42
7	242.65	41060.7	0.5895	312.27
8	236.15	35599.8	0.5252	308.06
9	229.65	30742.4	0.4663	303.79
10	223.15	26436.2	0.4127	299.46
11	216.65	22632.0	0.3639	295.06
12	216.65	19330.4	0.3108	295.06
13	216.65	16510.4	0.2655	295.06
14	216.65	14101.7	0.2268	295.06
15	216.65	12044.5	0.1937	295.06
16	216.65	10287.4	0.1654	295.06
17	216.65	8786.68	0.1413	295.06
18	216.65	7504.84	0.1207	295.06
19	216.65	6410.00	0.1031	295.06
20	216.65	5474.88	0.088	295.06

Gases Properties

Gas	M_{wt} kg/kmol	R J/kg-K	Cp J/kg-K	Cv J/kg-K	γ	h_f^o MJ/kmole
Air (O2+3.76N2)	29	287.00	1005.0	718.0	1.4	0
Oxygen	32	259.7	921.6	661.9	1.4	0
Carbon dioxide	44	188.92	841.8	652.9	1.289	-393.5
Nitrogen	28	296.80	1041.6	744.8	1.4	0
Water (gas)	18	461.52	1872.3	1410.8	1.327	-241.8

Liquid Fuel Property Data

Fuel	Density kg/m^3	LHV MJ/kg	LHV MJ/litre
Avgas	708.2	43.7	30.9
Jet A1	793.0	43.1	34.2

Planetary Data for the Earth

Quantity	Value
Gravity	$9.81 m/s^2$
Mean Gas Constant	287J/kg-K

Conversion and Reference Data

1Knot = 1.85kph Reference Temperature 298K R_u=8.314 kJ/kmol.K

***Equation Sheet:* state condition numbering used in equations is arbitrary.**

$$S = bc \qquad A = \frac{b}{c} \qquad \mathrm{V_{cal}} = \mathrm{V_E} = \mathrm{V_0}\sqrt{\frac{\rho}{\rho_s}} = \mathrm{V_0}\sqrt{\sigma}$$

$$L = \frac{1}{2}\rho U^2 c_L S \qquad D = \frac{1}{2}\rho U^2 c_D S \qquad c_L = a\alpha \qquad \eta_p = \frac{1}{1+w/V}$$

$$\dot{W}_I = (IMEP)\dot{V}_s \qquad \dot{V}_s = A_P L_S \frac{N}{2} n \qquad \eta = \frac{\dot{W}_n}{\dot{Q}_H} \qquad \dot{m} = \rho V A$$

$$TSFC = \frac{\dot{m}_f}{F_N} \qquad BSFC = \frac{\dot{m}_f}{\dot{W}_b} \qquad F_s = \frac{F_N}{\dot{m}_a} \qquad f = \frac{\dot{m}_f}{\dot{m}_a}$$

$$\dot{W}_b = F_T(w+u) \qquad w = \frac{1}{2}\left[\left(u^2 + \frac{2F_T}{\rho A_p}\right)^{1/2} - u\right] \qquad \dot{W}_{b,o} = F_{To}{}^{3/2}/\sqrt{2\rho A_p}$$

$$p = \rho RT \qquad pv = RT \qquad \gamma = C_p/C_v \qquad R = C_p - C_v$$

$$(pv^\gamma)_1 = (pv^\gamma)_2 \qquad (Tv^{\gamma-1})_1 = (Tv^{\gamma-1})_2$$

$$\sum Q + \sum W = 0 \qquad Q + W = \Delta U \qquad LHV = \left[(h_f^o)_R - (h_f^o)_P\right]/M_{wf}$$

$$\left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_1 = \left(\frac{p}{\rho} + u + \frac{C^2}{2}\right)_2 + Q + W$$

$$\left(\frac{p}{\rho} + u\right) = h \qquad (h_2 - h_1) = C_p(T_2 - T_1) \qquad h = C_p(T - T_R)$$

$$\frac{p_0}{p} = \left(\frac{T_0}{T}\right)^{\frac{\gamma}{\gamma-1}} \qquad \frac{p_{02}}{p_{01}} = \left(\frac{T_{02'}}{T_{01}}\right)^{\frac{\gamma}{\gamma-1}} \qquad \eta_i = \frac{T_{01'} - T_a}{T_{01} - T_a}$$

$$T_{0a} = T_a + \frac{C_a^2}{2c_p} \qquad \frac{p_{01}}{p_a} = \left[1 + \eta_i \frac{C_a^2}{2c_p T_a}\right]^{\frac{\gamma_a}{\gamma_a - 1}} \qquad TP = \dot{W}_s \eta_p + C_a F_N \qquad \eta_c = \frac{T_{02'} - T_{01}}{T_{02} - T_{01}}$$

$$\eta_t = \frac{T_{03} - T_{04}}{T_{03} - T_{04'}} \qquad \eta_j = \frac{T_{04} - T_5}{T_{04} - T_{5'}} \qquad \frac{p_{04}}{p_c} = \left[1 - \frac{1}{\eta_j}\left(\frac{\gamma-1}{\gamma+1}\right)\right]^{-\left(\frac{\gamma}{\gamma-1}\right)} \qquad \frac{T_c}{T_{04}} = \left(\frac{2}{\gamma+1}\right)$$

$$M_{n,1} = M_1 \sin\beta \qquad M_2 = M_{n,2}/\sin(\beta - \theta)$$

$$a = \sqrt{R\gamma T_c} \qquad M = \frac{C}{a} \qquad B = \frac{\dot{m}_c}{\dot{m}_h} \qquad FPR = p_{02}/p_{01} \quad OPR = p_{03}/p_{01}$$

$$C_J = \left[\left(\frac{2\gamma}{\gamma-1}\right)RT_c\left(1-\left(\frac{p_e}{p_c}\right)^{(\gamma-1)/\gamma}\right)\right]^{1/2} \qquad \frac{A}{A^*} = \frac{1}{M}\left(\frac{2+(\gamma-1)M^2}{\gamma+1}\right)^{(\gamma+1)/(2\gamma-2)}$$

$$\Delta C = C_e \ln(M_R) - gt_b \sin\theta \qquad z_b = C_o t_o + C_e t_b\left(\frac{1}{M_R - 1}\ln\left(\frac{1}{M_R}\right)+1\right) - \frac{gt_b^2 \sin\theta}{2}$$

$$z_c = z_b + \frac{r_o C_b^2}{2g_o r_o - C_b^2} \qquad M_R = \frac{M_o}{M_f} \qquad C_{esc} = r_o\sqrt{\left(\frac{2g_o}{r_o+z}\right)} \qquad C_{sat} = r_o\sqrt{\left(\frac{g_o}{r_o+z}\right)}$$

Normal Shock Tables

TABLE E.4 Normal Shock Flow of an Ideal Gas ($k = 1.4$)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.00	1.00000	1.00000	1.00000	1.00000	1.00000	1.8929	1.49	0.70466	2.4234	1.3136	1.8449	0.93292	3.3756
1.01	0.99013	1.02345	1.00665	1.01669	0.99999	1.9152	1.50	0.70109	2.4583	1.3202	1.8621	0.92978	3.4133
1.02	0.98052	1.04713	1.01325	1.03344	0.99998	1.9379	1.51	0.69758	2.4934	1.3269	1.8792	0.92658	3.4512
1.03	0.97115	1.07105	1.01981	1.05024	0.99997	1.9610	1.52	0.69413	2.5288	1.3336	1.8962	0.92331	3.4894
1.04	0.96202	1.09520	1.02634	1.06709	0.99994	1.9845	1.53	0.69073	2.5644	1.3403	1.9133	0.91999	3.5279
1.05	0.95312	1.1196	1.03284	1.08398	0.99987	2.0083	1.54	0.68739	2.6003	1.3470	1.9303	0.91662	3.5667
1.06	0.94444	1.1442	1.03931	1.10092	0.99976	2.0325	1.55	0.68410	2.6363	1.3538	1.9473	0.91319	3.6058
1.07	0.93598	1.1690	1.04575	1.11790	0.99962	2.0570	1.56	0.68086	2.6725	1.3606	1.9643	0.90970	3.6451
1.08	0.92772	1.1941	1.05217	1.13492	0.99944	2.0819	1.57	0.67768	2.7090	1.3674	1.9812	0.90615	3.6847
1.09	0.91965	1.2194	1.05856	1.15199	0.99921	2.1072	1.58	0.67455	2.7458	1.3742	1.9981	0.90255	3.7245
1.10	0.91177	1.2450	1.06494	1.1691	0.99892	2.1328	1.59	0.67147	2.7828	1.3811	2.0149	0.89889	3.7645
1.11	0.90408	1.2708	1.07130	1.1862	0.99858	2.1588	1.60	0.66844	2.8201	1.3880	2.0317	0.89520	3.8049
1.12	0.89656	1.2968	1.07764	1.2034	0.99820	2.1851	1.61	0.66545	2.8575	1.3949	2.0485	0.89144	3.8456
1.13	0.88922	1.3230	1.08396	1.2206	0.99776	2.2118	1.62	0.66251	2.8951	1.4018	2.0652	0.88764	3.8866
1.14	0.88204	1.3495	1.09027	1.2378	0.99726	2.2388	1.63	0.65962	2.9330	1.4088	2.0820	0.88380	3.9278
1.15	0.87502	1.3762	1.09657	1.2550	0.99669	2.2661	1.64	0.65677	2.9712	1.4152	2.0986	0.87992	3.9693
1.16	0.86816	1.4032	1.10287	1.2723	0.99605	2.2937	1.65	0.65396	3.0096	1.4228	2.1152	0.87598	4.0111
1.17	0.86145	1.4304	1.10916	1.2896	0.99534	2.3217	1.66	0.65119	3.0482	1.4298	2.1318	0.87201	4.0531
1.18	0.85488	1.4578	1.11544	1.3069	0.99455	2.3499	1.67	0.64847	3.0870	1.4369	2.1484	0.86800	4.0954
1.19	0.84846	1.4854	1.12172	1.3243	0.99371	2.3786	1.68	0.64579	3.1261	1.4440	2.1649	0.86396	4.1379
1.20	0.84217	1.5133	1.1280	1.3416	0.99280	2.4075	1.69	0.64315	3.1654	1.4512	2.1813	0.85987	4.1807
1.21	0.83601	1.5414	1.1343	1.3590	0.99180	2.4367	1.70	0.64055	3.2050	1.4583	2.1977	0.85573	4.2238
1.22	0.82998	1.5698	1.1405	1.3764	0.99073	2.4662	1.71	0.63798	3.2448	1.4655	2.2141	0.85155	4.2672
1.23	0.82408	1.5984	1.1468	1.3938	0.98975	2.4961	1.72	0.63545	3.2848	1.4727	2.2304	0.84735	4.3108
1.24	0.81830	1.6272	1.1531	1.4112	0.98835	2.5263	1.73	0.63296	3.3250	1.4800	2.2467	0.84312	4.3547
1.25	0.81264	1.6562	1.1594	1.4286	0.98706	2.5568	1.74	0.63051	3.3655	1.4873	2.2629	0.83886	4.3989
1.26	0.80709	1.6855	1.1657	1.4460	0.98568	2.5876	1.75	0.62809	3.4062	1.4946	2.2791	0.83456	4.4433
1.27	0.80165	1.7150	1.1720	1.4634	0.98422	2.6187	1.76	0.62570	3.4472	1.5019	2.2952	0.83024	4.4880
1.28	0.79631	1.7448	1.1782	1.4808	0.98268	2.6500	1.77	0.62335	3.4884	1.5093	2.3113	0.82589	4.5330
1.29	0.79108	1.7748	1.1846	1.4983	0.98106	2.6816	1.78	0.62104	3.5298	1.5167	2.3273	0.82152	4.5783
1.30	0.78596	1.8050	1.1909	1.5157	0.97935	2.7135	1.79	0.61875	3.5714	1.5241	2.3433	0.81711	4.6238
1.31	0.78093	1.8354	1.1972	1.5331	0.97758	2.7457	1.80	0.61650	3.6133	1.5316	2.3592	0.81268	4.6695
1.32	0.77600	1.8661	1.2035	1.5505	0.97574	2.7783	1.81	0.61428	3.6554	1.5391	2.3751	0.80823	4.7155
1.33	0.77116	1.8970	1.2099	1.5680	0.97382	2.8112	1.82	0.61209	3.6978	1.5466	2.3909	0.80376	4.7618
1.34	0.76641	1.9282	1.2162	1.5854	0.97181	2.8444	1.83	0.60993	3.7404	1.5542	2.4067	0.79926	4.8083
1.35	0.76175	1.9596	1.2226	1.6028	0.96972	2.8778	1.84	0.60780	3.7832	1.5617	2.4224	0.79474	4.8551
1.36	0.75718	1.9912	1.2290	1.6202	0.96756	2.9115	1.85	0.60570	3.8262	1.5694	2.4381	0.79021	4.9022
1.37	0.75269	2.0230	1.2354	1.6376	0.96534	2.9455	1.86	0.60363	3.8695	1.5770	2.4537	0.78567	4.9498
1.38	0.74828	2.0551	1.2418	1.6550	0.96304	2.9798	1.87	0.60159	3.9130	1.5847	2.4693	0.78112	4.9974
1.39	0.74396	2.0874	1.2482	1.6723	0.96065	3.0144	1.88	0.59957	3.9568	1.5924	2.4848	0.77656	5.0453
1.40	0.73971	2.1200	1.2547	1.6896	0.95819	3.0493	1.89	0.59758	4.0008	1.6001	2.5003	0.77197	5.0934
1.41	0.73554	2.1528	1.2612	1.7070	0.95566	3.0844	1.90	0.59562	4.0450	1.6079	2.5157	0.76735	5.1417
1.42	0.73144	2.1858	1.2676	1.7243	0.95306	3.1198	1.91	0.59368	4.0894	1.6157	2.5310	0.76273	5.1904
1.43	0.72741	2.2190	1.2742	1.7416	0.95039	3.1555	1.92	0.59177	4.1341	1.6236	2.5463	0.75812	5.2394
1.44	0.72345	2.2525	1.2807	1.7589	0.94765	3.1915	1.93	0.58988	4.1790	1.6314	2.5615	0.75347	5.2886
1.45	0.71956	2.2862	1.2872	1.7761	0.94483	3.2278	1.94	0.58802	4.2242	1.6394	2.5767	0.74883	5.3381
1.46	0.71574	2.3202	1.2938	1.7934	0.94196	3.2643	1.95	0.58618	4.2696	1.6473	2.5919	0.74418	5.3878
1.47	0.71198	2.3544	1.3004	1.8106	0.93901	3.3011	1.96	0.58437	4.3152	1.6553	2.6070	0.73954	5.4378
1.48	0.70829	2.3888	1.3070	1.8278	0.93600	3.3382	1.97	0.58258	4.3610	1.6633	2.6220	0.73487	5.4880

Normal Shock Tables Continued

TABLE E.4 (continued)

Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$	Ma_x	Ma_y	$\frac{p_y}{p_x}$	T_y/T_x	$\rho_y/\rho_x = V_x/V_y$	$\frac{p_{0,y}}{p_{0,x}}$	$p_{0,y}/p_x$
1.98	0.58081	4.4071	1.6713	2.6369	0.73021	5.5385	2.55	0.50831	7.4196	2.1875	3.3918	0.47927	8.8505
1.99	0.57907	4.4534	1.6794	2.6518	0.72554	5.5894	2.56	0.50740	7.4792	2.1976	3.4034	0.47540	8.9162
2.00	0.57735	4.5000	1.6875	2.6666	0.72088	5.6405	2.57	0.50651	7.5391	2.2077	3.4149	0.47155	8.9821
2.01	0.57565	4.5468	1.6956	2.6814	0.71619	5.6918	2.58	0.50562	7.5992	2.2179	3.4263	0.46772	9.0482
2.02	0.57397	4.5938	1.7038	2.6962	0.71152	5.7434	2.59	0.50474	7.6595	2.2281	3.4376	0.46391	9.1146
2.03	0.57231	4.6411	1.7120	2.7109	0.70686	5.7952	2.60	0.50387	7.7200	2.2383	3.4489	0.46012	9.1813
2.04	0.57068	4.6886	1.7203	2.7255	0.70218	5.8473	2.61	0.50301	7.7808	2.2486	3.4602	0.45636	9.2481
2.05	0.56907	4.7363	1.7286	2.7400	0.69752	5.8997	2.62	0.50216	7.8418	2.2589	3.4714	0.45262	9.3154
2.06	0.56747	4.7842	1.7369	2.7545	0.69284	5.9523	2.63	0.50132	7.9030	2.2693	3.4825	0.44891	9.3829
2.07	0.56589	4.8324	1.7452	2.7690	0.68817	6.0052	2.64	0.50048	7.9645	2.2797	3.4936	0.44522	9.4507
2.08	0.56433	4.8808	1.7536	2.7834	0.68351	6.0584	2.65	0.49965	8.0262	2.2901	3.5047	0.44155	9.5187
2.09	0.56280	4.9295	1.7620	2.7977	0.67886	6.1118	2.66	0.49883	8.0882	2.3006	3.5157	0.43791	9.5869
2.10	0.56128	4.9784	1.7704	2.8119	0.67422	6.1655	2.67	0.49802	8.1504	2.3111	3.5266	0.43429	9.6553
2.11	0.55987	5.0275	1.7789	2.8216	0.66957	6.2194	2.68	0.49722	8.2128	2.3217	3.5374	0.43070	9.7241
2.12	0.55830	5.0768	1.7874	2.8402	0.66492	6.2736	2.69	0.49642	8.2754	2.3323	3.5482	0.42713	9.7932
2.13	0.55683	5.1264	1.7960	2.8543	0.66029	6.3280	2.70	0.49563	8.3383	2.3429	3.5590	0.42359	9.8625
2.14	0.55538	5.1762	1.8046	2.8683	0.65567	6.3827	2.71	0.49485	8.4014	2.3536	3.5697	0.42007	9.9320
2.15	0.55395	5.2262	1.8132	2.8823	0.65105	6.4377	2.72	0.49408	8.4648	2.3643	3.5803	0.41657	10.002
2.16	0.55254	5.2765	1.8219	2.8962	0.64644	6.4929	2.73	0.49332	8.5284	2.3750	3.5909	0.41310	10.072
2.17	0.55114	5.3270	1.8306	2.9100	0.64185	6.5484	2.74	0.49256	8.5922	2.3858	3.6014	0.40965	10.142
2.18	0.54976	5.3778	1.8393	2.9238	0.63728	6.6042	2.75	0.49181	8.6562	2.3966	3.6119	0.40622	10.212
2.19	0.54841	5.4288	1.8481	2.9376	0.63270	6.6602	2.76	0.49107	8.7205	2.4074	3.6224	0.40282	10.283
2.20	0.54706	5.4800	1.8569	2.9512	0.62812	6.7163	2.77	0.49033	8.7850	2.4183	3.6328	0.39945	10.354
2.21	0.54572	5.5314	1.8657	2.9648	0.62358	6.7730	2.78	0.48960	8.8497	2.4292	3.6431	0.39610	10.426
2.22	0.54440	5.5831	1.8746	2.9783	0.61905	6.8299	2.79	0.48888	8.9147	2.4402	3.6533	0.39276	10.498
2.23	0.54310	5.6350	1.8835	2.9918	0.61453	6.8869	2.80	0.48817	8.9800	2.4512	3.6635	0.38946	10,569
2.24	0.54182	5.6872	1.8924	3.0052	0.61002	6.9442	2.81	0.48746	9.0454	2.4622	3.6737	0.38618	10.641
2.25	0.54055	5.7396	1.9014	3.0186	0.60554	7.0018	2.82	0.48676	9.1111	2.4733	3.6838	0.38293	10.714
2.26	0.53929	5.7922	1.9104	3.0319	0.60106	7.0597	2.83	0.48607	9.1770	2.4844	3.6939	0.37970	10.787
2.27	0.53805	5.8451	1.9194	3.0452	0.59659	7.1178	2.84	0.48538	9.2432	2.4955	3.7039	0.37649	10.860
2.28	0.53683	5.8982	1.9285	3.0584	0.59214	7.1762	2.85	0.48470	9.3096	2.5067	3.7139	0.37330	10.933
2.29	0.53561	5.9515	1.9376	3.0715	0.58772	7.2348	2.86	0.48402	9.3762	2.5179	3.7238	0.37013	11.006
2.30	0.53441	6.0050	1.9468	3.0846	0.58331	7.2937	2.87	0.48334	9.4431	2.5292	3.7336	0.36700	11.080
2.31	0.53322	6.0588	1.9560	3.0976	0.57891	7.3529	2.88	0.48268	9.5102	2.5405	3.7434	0.36389	11.154
2.32	0.53205	6.1128	1.9652	3.1105	0.57452	7.4123	2.89	0.48203	9.5775	2.5518	3.7532	0.36080	11.228
2.33	0.53089	6.1670	1.9745	3.1234	0.57015	7.4720	2.90	0.48138	9.6450	2.5632	3.7629	0.35773	11.302
2.34	0.52974	6.2215	1.9838	3.1362	0.56580	7.5319	2.91	0.48074	9.7127	2.5746	3.7725	0.35469	11.377
2.35	0.52861	6.2762	1.9931	3.1490	0.56148	7.5920	2.92	0.48010	9.7808	2.5860	3.7821	0.35167	11.452
2.36	0.52749	6.3312	2.0025	3.1617	0.55717	7.6524	2.93	0.47946	9.8491	2.5975	3.7917	0.34867	11.527
2.37	0.52638	6.3864	2.0119	3.1743	0.55288	7.7131	2.94	0.47883	9.9176	2.6090	3.8012	0.34570	11.603
2.38	0.52528	6.4418	2.0213	3.1869	0.54862	7.7741	2.95	0.47821	9.986	2.6206	3.8106	0.34275	11.679
2.39	0.52419	6.4974	2.0308	3.1994	0.54438	7.8354	2.96	0.47760	10.055	2.6322	3.8200	0.33982	11.755
2.40	0.52312	6.5533	2.0403	3.2119	0.54015	7.8969	2.97	0.47699	10.124	2.6438	3.8294	0.33692	11.831
2.41	0.52206	6.6094	2.0499	3.2243	0.53594	7.9587	2.98	0.47638	10.194	2.6555	3.8387	0.33404	11.907
2.42	0.52100	6.6658	2.0595	3.2366	0.53175	8.0207	2.99	0.47578	10.263	2.6672	3.8479	0.33118	11.984
2.43	0.51996	6.7224	2.0691	3.2489	0.52758	8.0830	3.00	0.47519	10.333	2.6790	3.8571	0.32834	12.061
2.44	0.51894	6.7792	2.0788	3.2611	0.52344	8.1455	3.50	0.45115	14.125	3.3150	4.2608	0.21295	16.242
2.45	0.51792	6.8362	2.0885	3.2733	0.51932	8.2083	4.00	0.43496	18.500	4.0469	4.5714	0.13876	21.068
2.46	0.51691	6.8935	2.0982	3.2854	0.51521	8.2714	4.50	0.42355	23.458	4.8751	4.8119	0.09170	26.539
2.47	0.51592	6.9510	2.1080	3.2975	0.51112	8.3347	5.00	0.41523	29.000	5.8000	5.0000	0.06172	32.654
2.48	0.51493	7.0088	2.1178	3.3095	0.50706	8.3983	6.00	0.40416	41.833	7.941	5.2683	0.02965	46.815
2.49	0.51395	7.0668	2.1276	3.3214	0.50303	8.4622	7.00	0.39736	57.000	10.469	5.4444	0.01535	63.552
2.50	0.51299	7.1250	2.1375	3.3333	0.49902	8.5262	8.00	0.39289	74.500	13.387	5.5652	0.00849	82.865
2.51	0.51204	7.1834	2.1474	3.3451	0.49502	8.5904	9.00	0.38980	94.333	16.693	5.6512	0.00496	104.753
2.52	0.51109	7.2421	2.1574	3.3569	0.49104	8.6549	10.00	0.38757	116.50	20.388	5.7143	0.00304	129.217
2.53	0.51015	7.3010	2.1674	3.3686	0.48709	8.7198	∞	0.37796	∞	∞	6.000	0	∞
2.54	0.50923	7.3602	2.1774	3.3802	0.48317	8.7850							

$$M_{n,1} = M_1 \sin\beta \qquad M_2 = M_{n,2} / \sin(\beta - \theta)$$

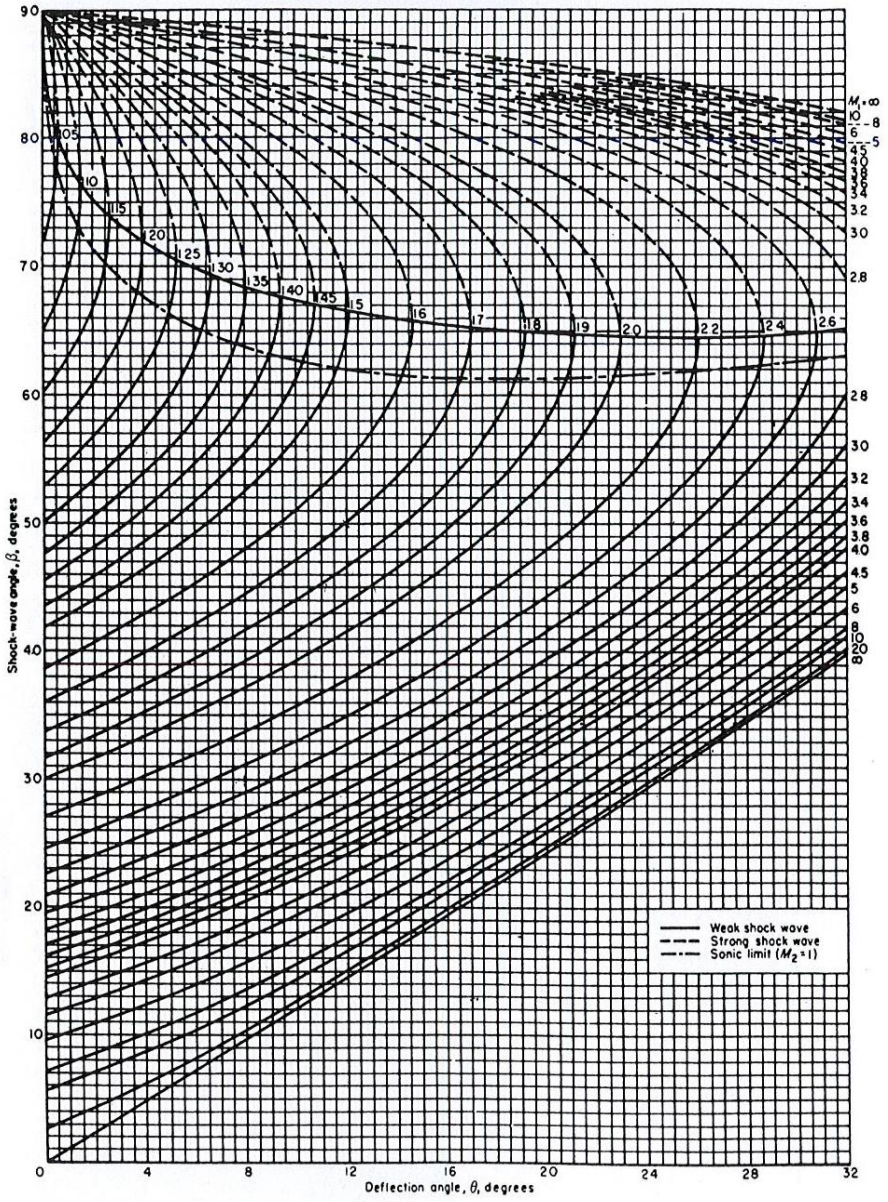


FIGURE 9.7
Oblique shock properties: $\gamma = 1.4$. The θ-β-M diagram. (*From NACA Report 1135, Ames Research Staff, "Equations, Tables and Charts for Compressible Flow," 1953.*)